

結合3D CAD合成影像與幾何特徵之工業零件細粒度分類與長度估算框架研究

A Framework for Fine-Grained Classification and Length Estimation of Industrial Parts Combining Synthetic 3D CAD Images and Geometric Features

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Outline

- Introduction
- Related works
- Research Method
- Experiments
- Conclusion

Introduction

Background

- A single machine may consist of hundreds of components, finding the right components can be time consuming. **Many components differ only in length or hole size**, making it hard for workers to identify the correct part without experience.
- **Deep learning models** are powerful, it will be **helpful for recognition**.
- Since recognition and assembly happen at the same time, **AR helmets can be used to capture image for deep learning model to recognize**.



Class 39

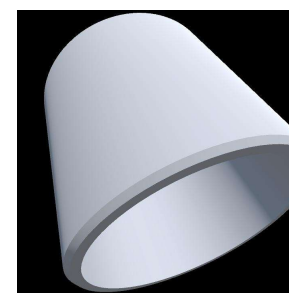
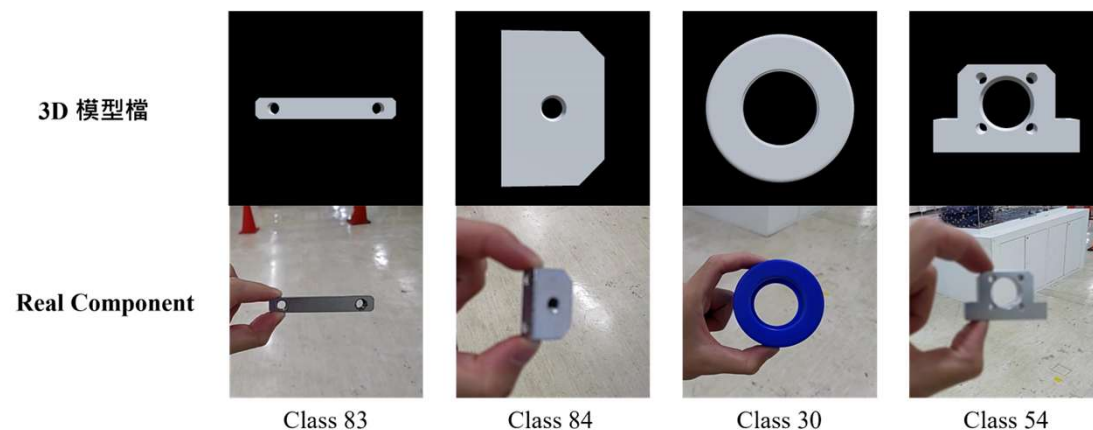


Class 102



Background

- The machines produced in the factory are assembled based on customers' orders, it's **difficult to obtain real-components in advance** and take a large number of real images to train deep learning model.
- Using **3D component rendered images as training data** to train a deep learning model is relatively feasible solution.
- The **3D model files** provided by the factory **do not contain texture information**.



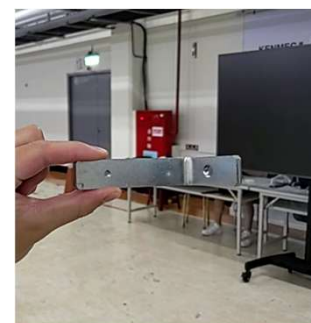
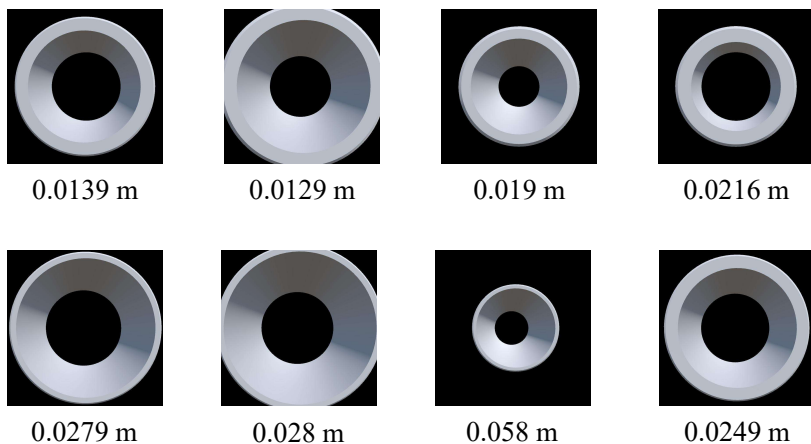
untextured



textured

Motivation

- Some of industrial components are similar and some of which are **only slightly different in length**.
- In the existing method, **Fine-grained classification model** cannot recognize industrial components that differ only in length. **Incorporating length information** may improve classification performance.
- In the inference phase, several angles for industrial components contains different information, the angles for length measurement might not suitable for recognition.



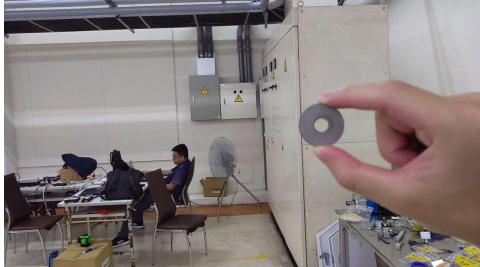
Recognition



Length Estimation

Motivation

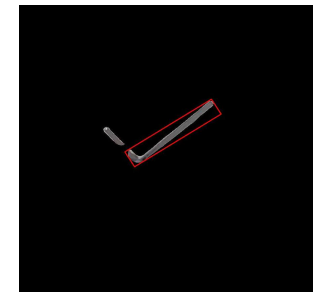
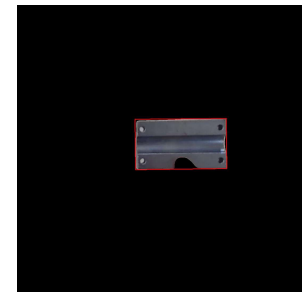
- AR helmets is equipped with depth camera, but the **RGB and Depth images have different resolution**, object localization seems to be difficult.
- In real-world scenarios, captured images **often include unrelated factors** (such as stacked metal parts, cluttered floors, etc.), so the heat map isn't focus on the correct position.
- The current method (related work) for deciding the longest side of industrial components has not considered the characteristics of their geometry.



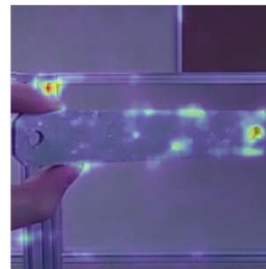
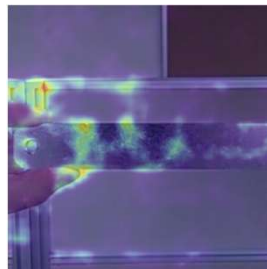
RGB



Depth



Heat map



Problem Definition

- When the data in training phase is the untextured 3D CAD rendered images, and the data used in inference phase is RGB-D images of real-world industrial components with complex backgrounds, how to build fine-grained classification framework without using real-images for training, considering device constraints and difference in viewing angle for both classification and length estimation?

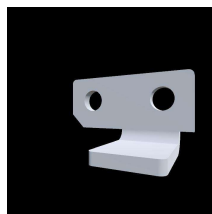
Equipment



Microsoft HoloLens2

Input Data

(Training)



(Testing)



Output Data

Class	Top1-Acc	Top5-Acc
Class XX	75.00%	100.00%
Class XX	50.00%	95.00%
Class XX	25.00%	50.00%

Objective & Contribution

- **Objective:**

Address the **challenges for classification in fine-grained components and length estimation** with different shapes in real-world scenarios.

- **Contribution:**

1. Adopting Two-stage identification procedure with background removal module and shape-based length estimation into 3D-industrial components classification without using real-images in training, **compared with existing method can outperform 11.84% accuracy.**
2. The purpose of **background removal module** in classification and length estimation suppressed the background interference, **improving classification accuracy by 2% and reducing length error by 1.1m.**
3. And the purposed of **shape-based length estimation module** is considering the geometrics of the components that can calculate the accurate longest side of the components, **reducing length error by 1cm.**

Related works

Industrial Components Recognition

- [1] : Utilizing pretrained weights, such as those from ImageNet, can improve model performance.
- [4] : Studies have shown that deep learning models outperform traditional CNN-based methods in industrial components recognition
- [2] [4] : Incorporating length information as an auxiliary feature can be further enhance the performance of recognition.
- [3] : Microsoft HoloLens 2 is equipped with an RGB camera and a depth sensor, making it suitable for both recognition and length estimation tasks.

[1] J. Krueger, J. Lehr, M. Schlueter, and N. Bischoff, "Deep learning for part identification based on inherent features," *CIRP Annals*, vol. 68, no. 1, pp. 9–12, 2019.

[2] J. Lehr, M. Schlüter, and J. Krüger, "Classification of similar objects of different sizes using a reference object by means of convolutional neural networks," in *2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, pp. 1519–1522, IEEE, 2019.

[3] 程劭予, "用於輔助工業零件辨識之尺寸估算系統," 碩士論文, 國立中央大學, 2024. 臺灣博碩士論文知識加值系統.

[4] 吳明憲, "使用無紋理之 3D CAD 工業零件模型結合長度檢測實現細粒度真實工業零件影像分類," 碩士論文, 國立中央大學, 2024. 臺灣博碩士論文知識加值系統.

Domain Shift and Domain Adaptation

- [5][6] : Most studies on domain shift or domain adaptation use 3D CAD models that contains texture information, which differs from the untextured 3D CAD models in our dataset.
- [5][6] : Synthetic images are pre-processed through various images pre-processing to make them resemble real-world images. Ex. adding background 、 horizontal flip.
- During the training phase, the Unsupervised Domain Adaptation Pipeline (UDA pipeline) [5] adopts CDAN-MCC methods to align features between synthetic and real images.

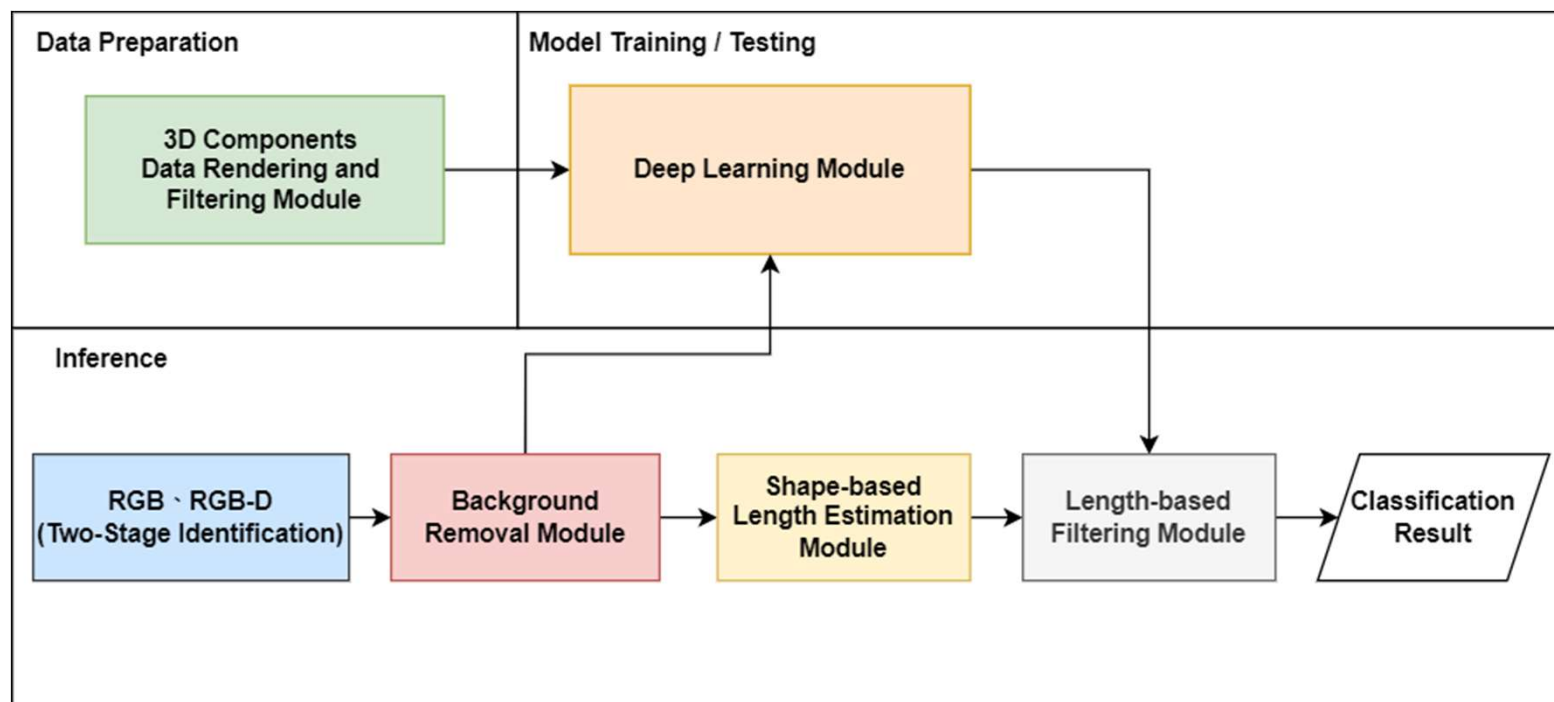
[5] D. Ritter, M. Hemberger, M. Hönig, V. Stopp, E. Rodner, and K. Hildebrand, “Cad models to real-world images: A practical approach to unsupervised domain adaptation in industrial object classification,” in Joint European Conference on Machine Learning and Knowledge Discovery in Databases, pp. 399–415, Springer, 2023.

[6] X. Zhu, T. Bilal, P. Mårtensson, L. Hanson, M. Björkman, and A. Maki, “Towards sim-to-real industrial parts classification with synthetic dataset,” in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 4454–4463, 2023.

Research Method

System Architecture

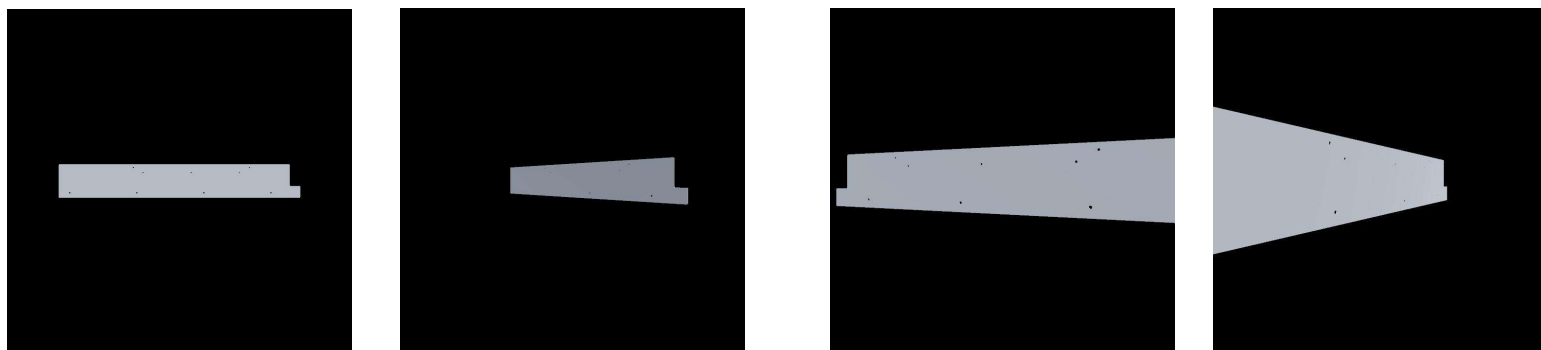
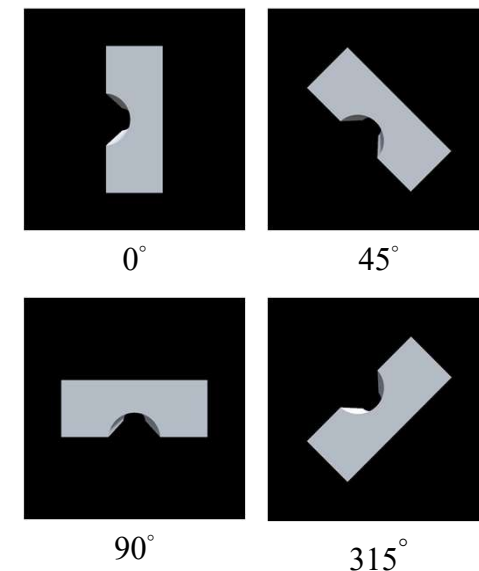
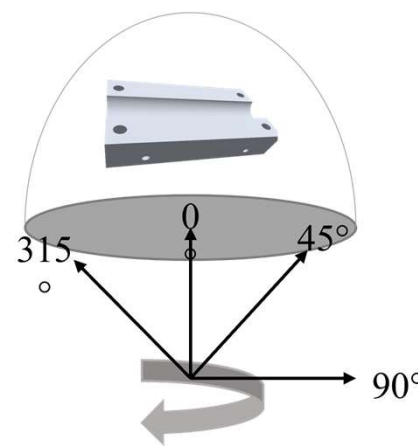
- Divide the system into three parts :
 1. Data Preparation
 2. Model (Training / Testing)
 3. Inference



3D Components Data Rendering and Filtering Module

1. 3D Components Data Rendering

- **Purpose :**
 - Deal with scenarios that require a large number of images for deep learning model training.
- **Method :**
 - Select an angle (0° , 45° , 90° , 315°) along the y-axis and **rotate the components to generate images.**
 - For component's real length $\geq 1\text{m}$, will **adjust the distance** to capture the more subtle features.



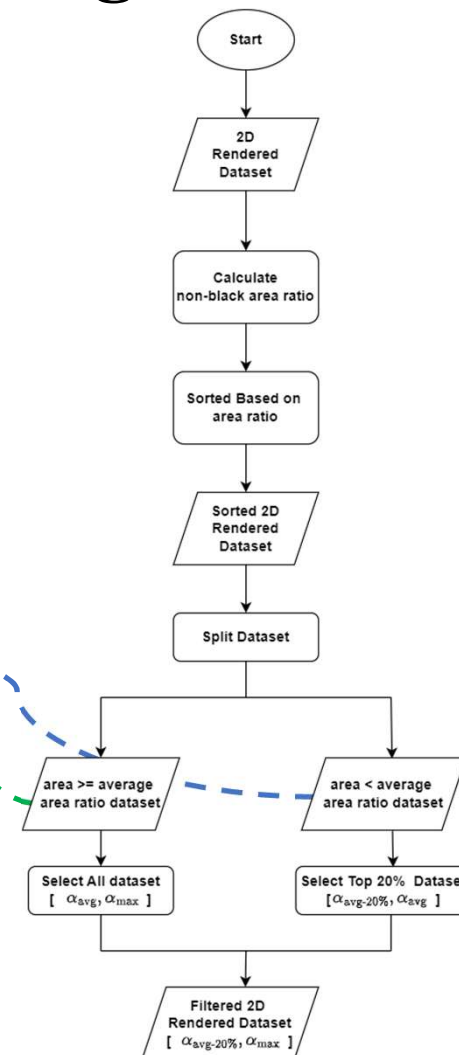
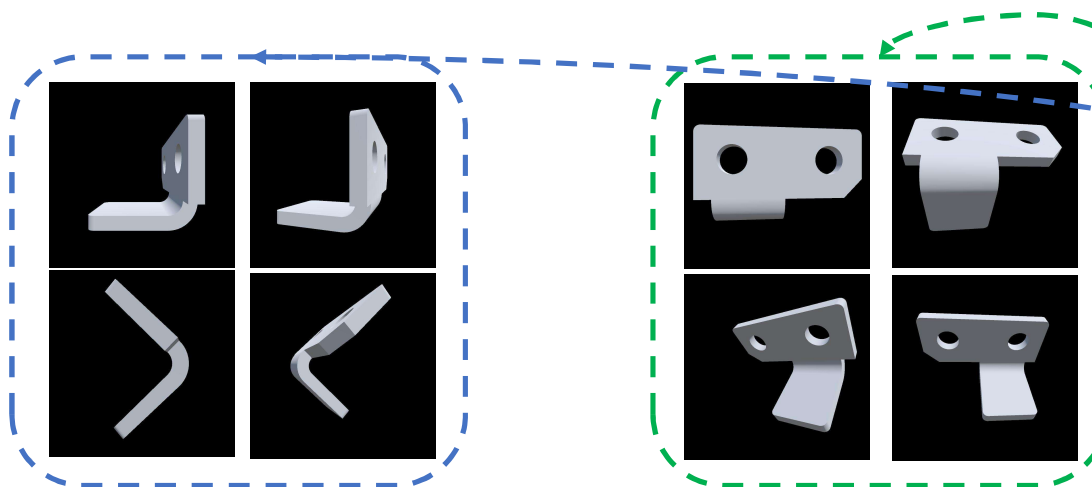
• Before Adjustment

• After Adjustment

3D Components Data Rendering and Filtering Module

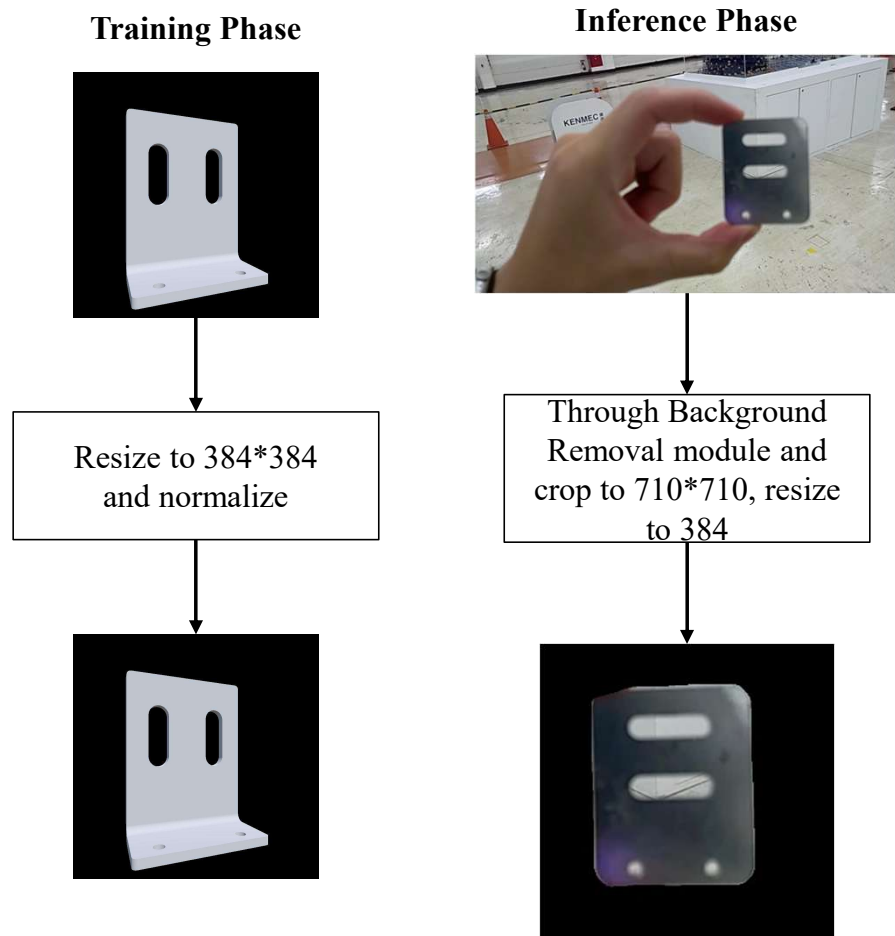
2. Dataset filtering

- **Purpose :**
 - Remove images that lack distinctive features may confuse the model.
- **Method :**
 - Since the background of the rendered images is black, the usefulness of an image can be determined by **calculating the area ratio of the components.**



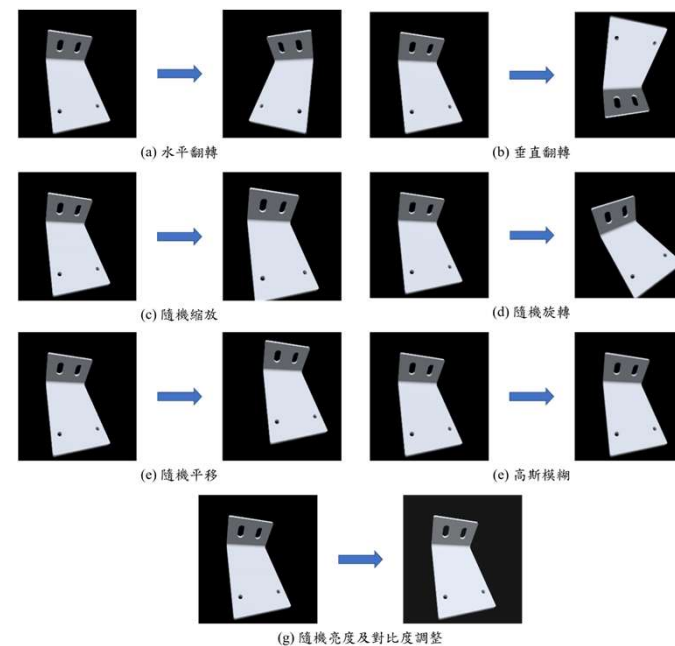
Deep Learning Module

• Data Preprocessing



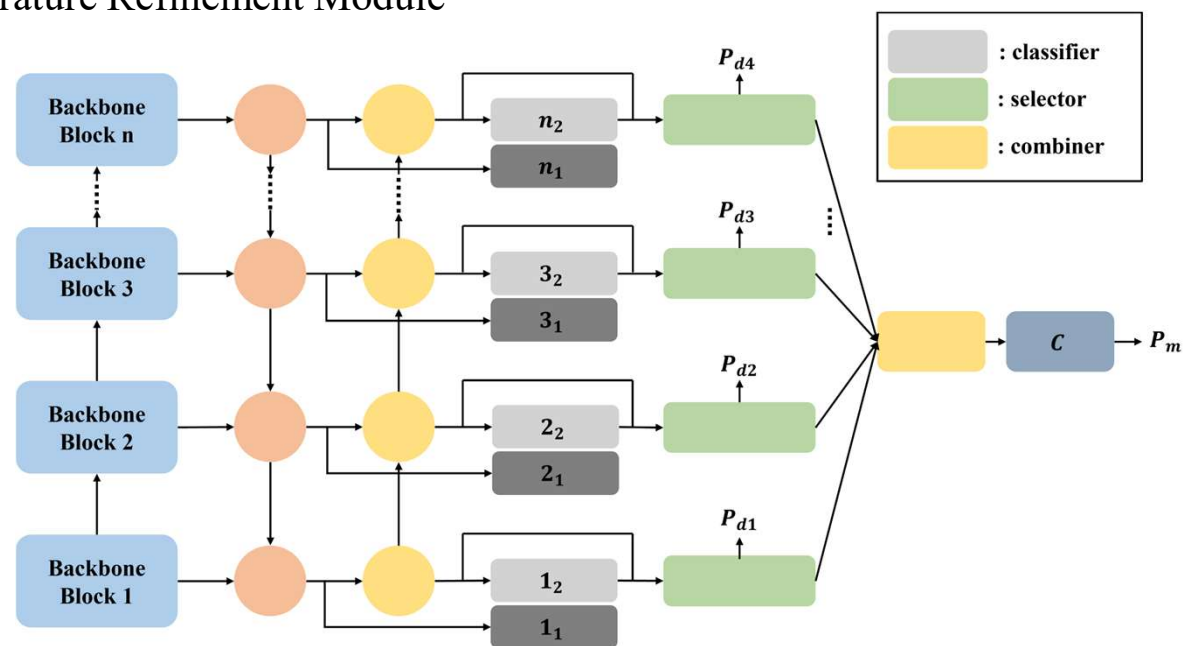
• Data Augmentation :

- Random horizontal flip
- Random vertical flip
- Random rotate
- Random Shift
- Random scale
- Random brightness contrast
- Random Gaussian Blur



Deep Learning Module

- **HERBS** model is the main classification model for the framework.
- Backbone : Swin-Transformer Large
- Its mains two module :
 1. Background Suppression Module
 2. High-temperature Refinement Module



- model architecture

Two-Stage Identification Procedure

- **Purpose :**
 - Based on the images from different angles, length estimation and model classification are fully utilizing the components information from different angles and reduce the missing features or estimated error that may caused by a single images
- **Method :**
 - In inference phase, two images are input from different angles and shooting strategy.
 - 1. **RGB** : Focus on capturing **fine-grained features of the components** for deep learning model **recognition**.
 - 2. **RGB-D** : To prevent depth camera cannot receiving the depth information correctly, the distance need to be 40 cm away from the camera. And captured the angles that for the **component longest side**.



RGB

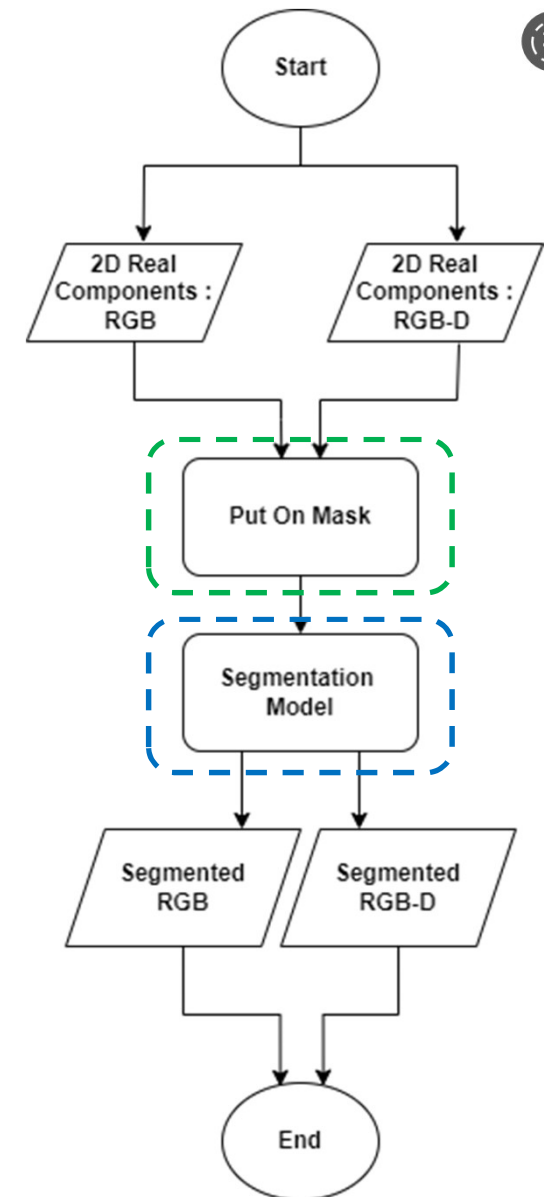
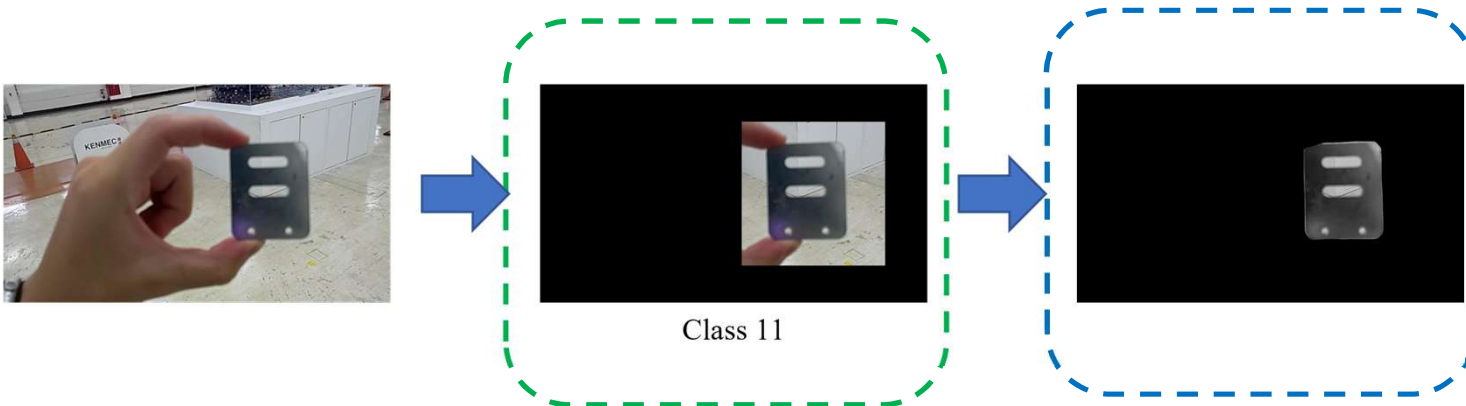


RGB-D



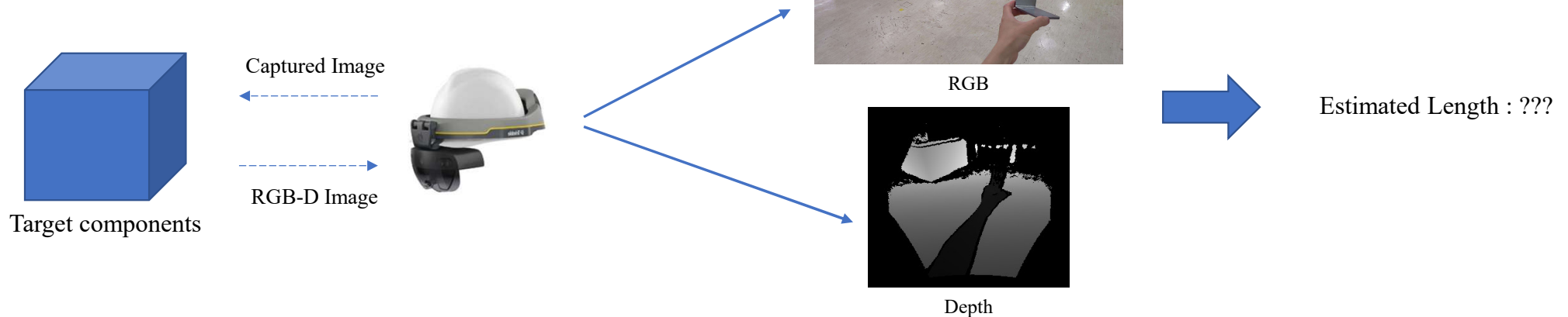
Background Removal Module

- **Purpose :**
 - To **reduce the interference from environmental factors** during model recognition and minimize **the impact of background limitations** in length estimation.
- **Method :**
 - By incorporating a segmentation model (FastSAM), text prompts are used to automatically identify and separate target items.



Shape-Based Length Estimation Module

- **Purpose :**
 - Calculate the length of the target component from RGB-D image
- **Method :**
 - Including five steps to calculate the length of target components
 1. Finding target component contours.
 2. Using OpenCV to find the target component's position.
 3. Mapping pixel position to depth image.
 4. Get the depth value of the component.
 5. Calculate the length of the target component.

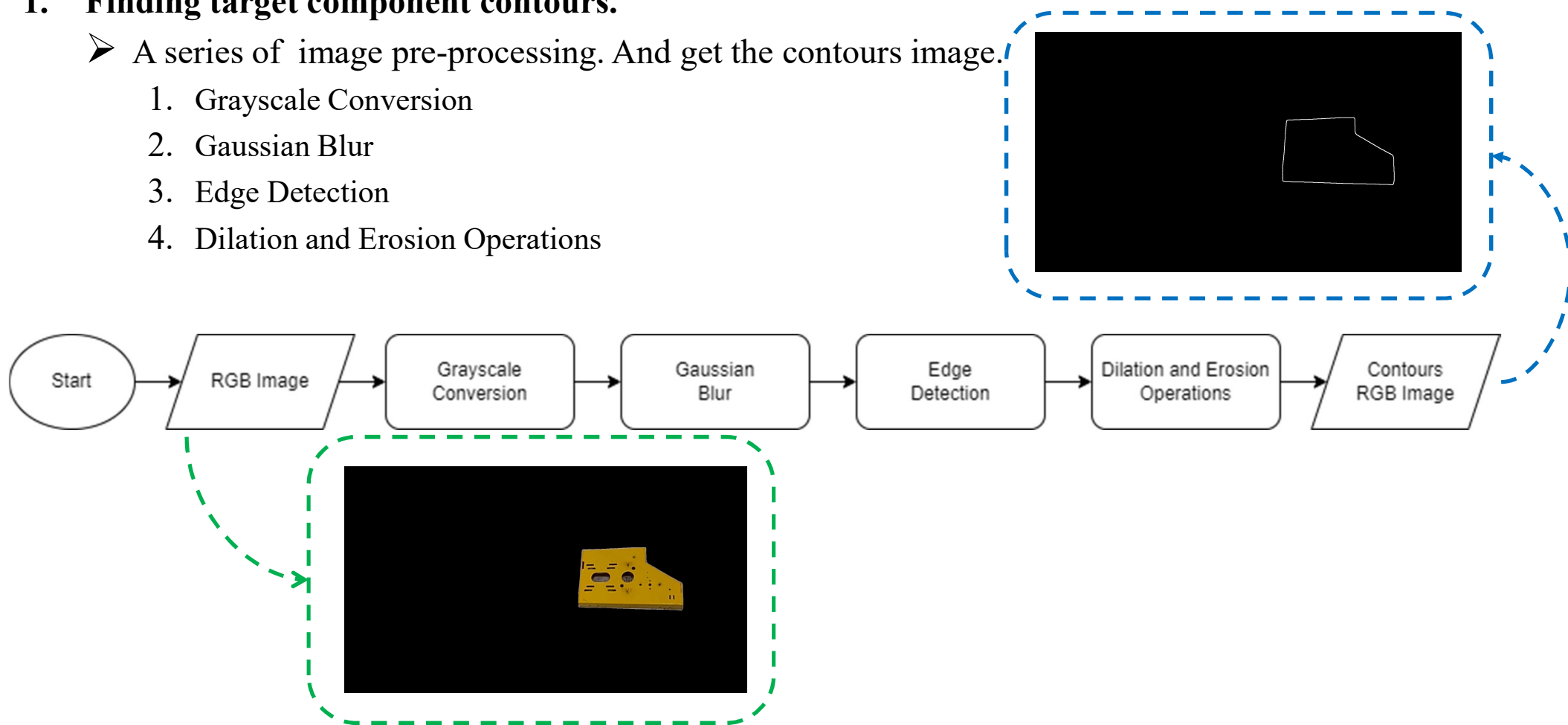


Shape-Based Length Estimate Module

1. Finding target component contours.

➤ A series of image pre-processing. And get the contours image.

1. Grayscale Conversion
2. Gaussian Blur
3. Edge Detection
4. Dilation and Erosion Operations

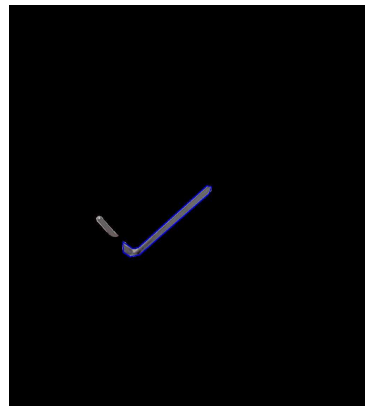
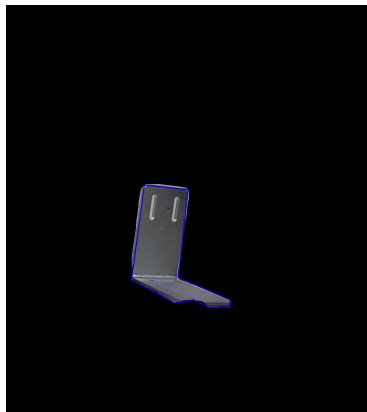


Shape-Based Length Estimate Module

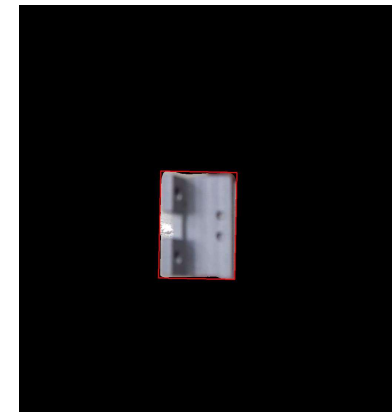
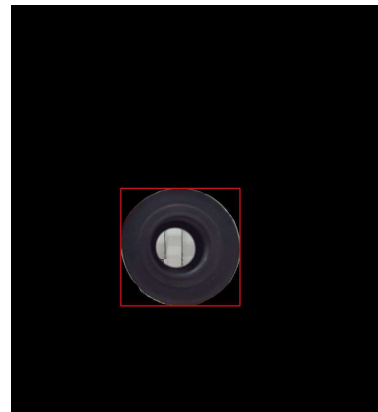
2. Using OpenCV to find the target component's position.

We obtain the contour images from the first step, and then classify the components into L-type and Non-L-type based on their geometric shapes.

- For **L-type components**, OpenCV's `arcLength()` and `approxPolyDP()` are used to extract the vertices of the contour and draw a shape that closely matches the actual contour.
- For **Non-L-type components**, OpenCV's `minAreaRect()` is used to find the smallest bounding rectangle that can enclose the component.



• L-type Components



• Non L-type Components

Shape-Based Length Estimate Module

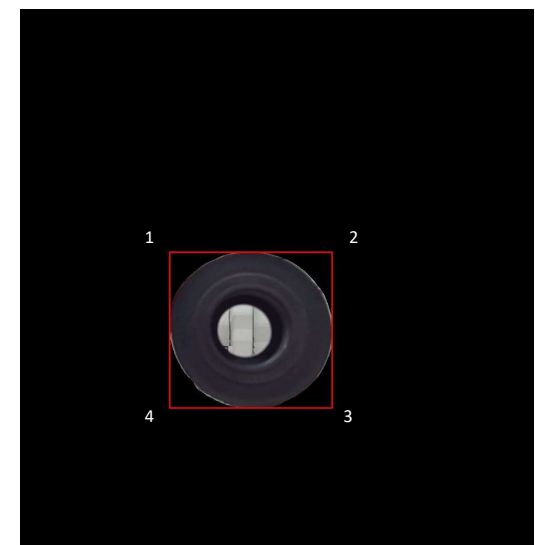
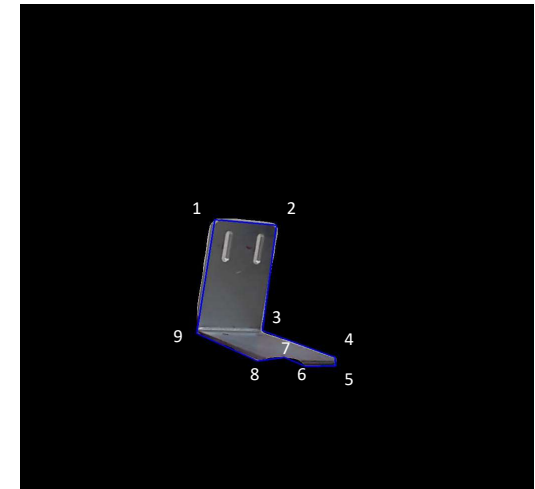
3. Mapping pixel position to depth image.

- Utilizes Microsoft HoloLens 2 intrinsic and extrinsic 、 focal length 、 principal point 、 RGB and depth pose array.
- The process involves transformations between the following coordinate systems:
 1. Pixel Coordinate System
 2. Image Coordinate System
 3. Camera Coordinate System
 4. World Coordinate System
- Main approach about the mapping :
 1. Depth images to World Coordinate system
 2. World Coordinate system to RGB Pixel Coordinate system

Shape-Based Length Estimate Module

4. Get the depth value of the component.

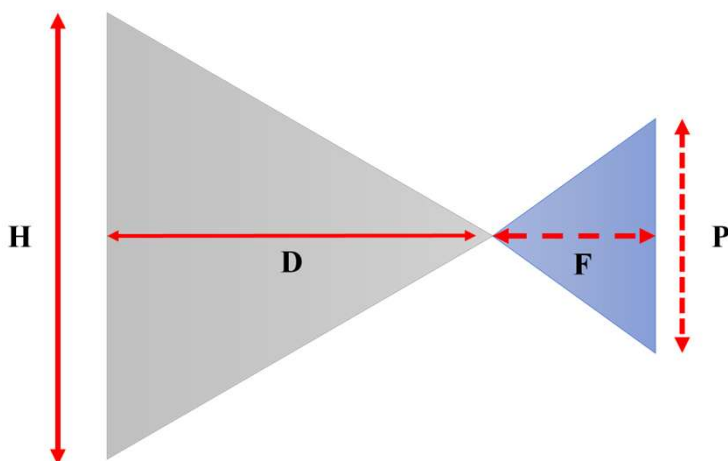
- L-type and Non-L-type components require different approaches for obtaining depth values.
- For L-type components:
 - The midpoint of the longest side can be taken as the depth value.
- For Non L-type components:
 - Take the midpoints of two long edge to form a line equation.
 - Traverse pixels along the line from one end of the longest edge.
 - Use the first valid (non-zero) depth value for estimation.



Shape-Based Length Estimate Module

5. Calculate the length of the target component.

- Get the target component longest side pixel length, depth value, and based on the similar formulas to get the real length of target components
- Pixel length (P) 、 Depth value (D) 、 Depth camera focal length (F) 、 Real length (H)



$$\frac{H}{P} = \frac{D}{F}$$

$$H = \frac{D * P}{F}$$

Length-Based Filtering Module

- **Purpose :**
 - Adjustment the prediction result from deep learning model based on estimated length.
- **Input :**
 1. Deep Learning Module prediction result (**Take Top10 prediction result**)
 2. Estimated Length
 3. Component's real length information
- Suppose Estimated Length = C , Tolerance Scope = T , and Component real length = P , then we check:

$$|C - P| \leq T(C), T(C) = \begin{cases} 0.055, & \text{if } C \geq 0.2659 \\ 0.0175, & \text{if } C < 0.2659 \end{cases}$$
- **Example :**
 - Prediction list = [34, 10, 6, 5, 8,]
 - Real length list = [0.3, 0.055, 0.1, 1.2, 1.33,]

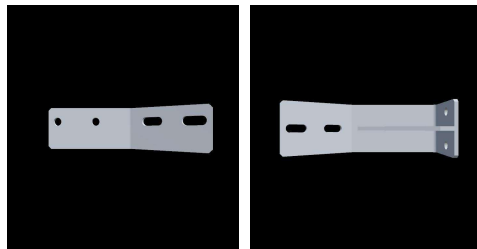
P

 - Estimated length (C) = 0.045
 - Tolerance scope (T) = 0.0175, because $C < 0.2659$.
$$|C - P| \leq T(C)$$
 - $|0.045 - 0.3| \leq 0.0175$
 - $|-0.255| \leq 0.0175$
 - $0.255 \leq 0.0175 \Rightarrow$ **False**
 - **Put Into last position**

Experiments

Dataset Introduction – 104 industrial components

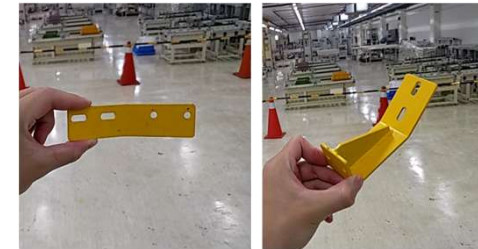
- Total Class : 104
- Training data is rendered from untextured 3D model and testing data is captured by Microsoft HoloLens 2
- Training Data : Synthetic RGB 、 Synthetic RGB-B



Synthetic RGB



Synthetic RGB-B



Testing data

Dataset	Number
Rendered Training Images	28383
Real Images For Training.	104
Testing Images (Length + Classification)	628 (304 + 324)

Dataset Introduction – topex-printer

- Total Class : 102
- Training data is rendered from textured 3D model
- Training Data : Topex-RGB 、 Topex-RGB-R 、 Topex-RGB-W
- Testing Data : Real images.

Training Data



Topex-RGB



Topex-RGB-R



Topex-RGB-W

Testing Data



	Number
Synthetic Training Images (Topex-RGB)	3264
Synthetic Training Images (Topex-RGB-R)	13056
Synthetic Training Images (Topex-RGB-W)	3264
Real Images For Training.	102
Testing Images (Length + Model)	6232 (429 + 5803)

Experiment 1: Best Accuracy

Objective	Compare our proposed method with UDA Pipeline
Description	- Evaluate accuracy results on 104 industrial components datasets, trained by 104 industrial components rendered images (Synthetic RGB) and 104 industrial components rendered added backgrounds images (Synthetic RGB-B) - Method Comparison: - Proposed Method - Related Work (UDA Pipeline) - The main solution provided by each method : - Proposed Method : Background Removal module 、 Length Estimation (Shape-based Length Estimation module + Length-based Filtering Module) - UDA Pipeline : Adding real images in training for transfer-learning (CDAN-MCC)
Metrics	Top1 accuracy 、 Top5 accuracy

	Proposed Method	UDA Pipeline
Adding real Images in training	✗	✓
Background Removal Module	✓	✗
Length Estimation	✓	✗

Dataset	Number
Rendered Training Images	28383
Real Images For Training.	104
Testing Images (Length + Classification)	628 (304 + 324)

Experiment 1: Result

- Compare proposed method and UDA Pipeline on 104 industrial components dataset.
- Experiment Setting :

	Proposed Method	UDA Pipeline
Adding real Images in training	✗	✓
Background Removal Module (BR)	✓	✗
Length Estimation (LE)	✓	✗

Train Dataset	Proposed Method (w/o adding real)		UDA Pipeline		Proposed Method (w/ adding real)		UDA Pipeline + BR + LE	
	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc
Synthetic RGB	60.72%	84.26%	48.88%	79.26%	67.16%	88.97%	60.96%	84.79%
Synthetic RGB-B	53.74%	74.60%	53.30%	78.85%	60.34%	83.00%	58.69%	83.00%

- Conclusion :
 - Even when the UDA pipeline add real images in the training phase cannot get better performance.
 - When all settings are the same, proposed method still performs better in terms of accuracy.

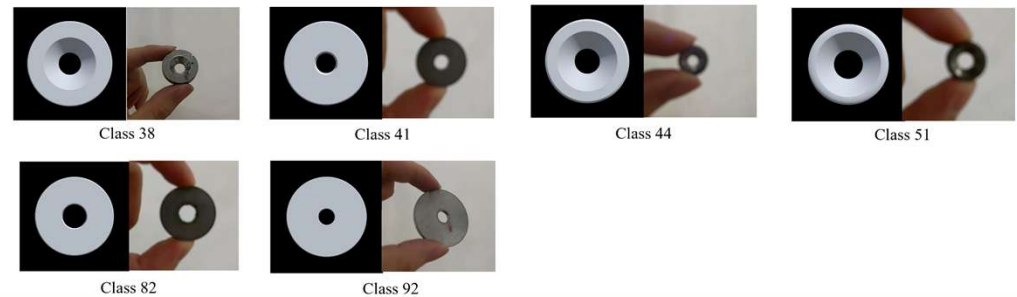
Experiment 2: Similar Components Performance

Objective	Evaluate the performance of our proposed method on similar components and compare with UDA Pipeline
Description	• The model trained by Synthetic RGB and Synthetic RGB-B dataset, same as Exp1. • Evaluate the result on similar components (四大混淆類群) 1. 端板群 (6 classes) 2. 套管群 (8 classes) 3. 輪子群 (8 classes) 4. 長方兩孔群 (3 classes) • Method Comparison: 1. Proposed Method 2. Related Work (UDA Pipeline)
Metrics	Top1 accuracy 、 Top5 accuracy

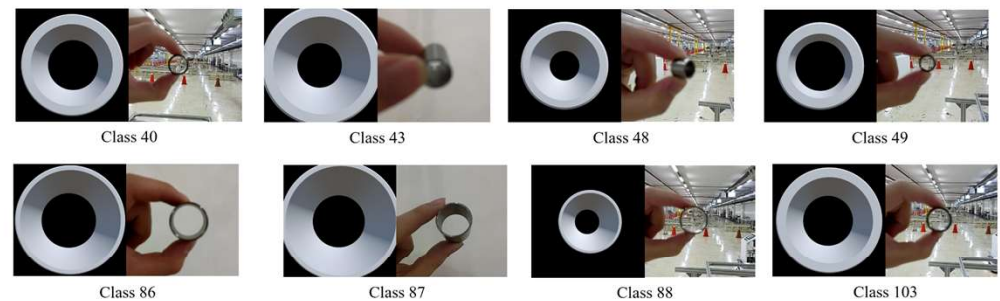
Similar Components

- Similar Components divide into 4 categories:
 1. 端板群 (6 classes)
 2. 套管群 (8 classes)
 3. 輪子群 (8 classes)
 4. 長方兩孔群 (3 classes)

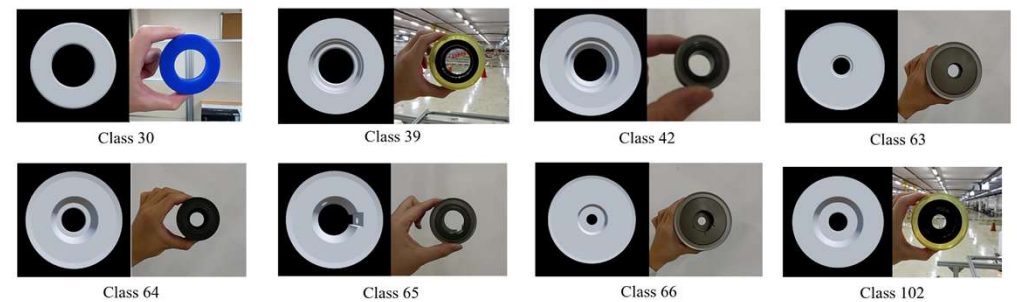
端板群



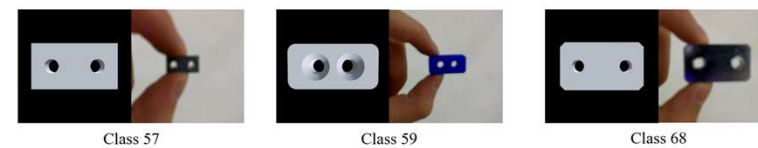
套管群



輪子群



長方兩孔群



Experiment 2 : Result

- Similar Components Total Result (Synthetic RGB)

Cluster	Proposed Method		UDA Pipeline	
	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc
端板群	57.33%	87.77%	5.56%	82.22%
套管群	22.5%	33.33%	8.33%	33.33%
輪子群	66.67%	86.7%	59.17%	87.50%
長方兩孔群	40%	60.00%	44.45%	77.78%
Total	46.63%	66.94%	29.38%	70.21%

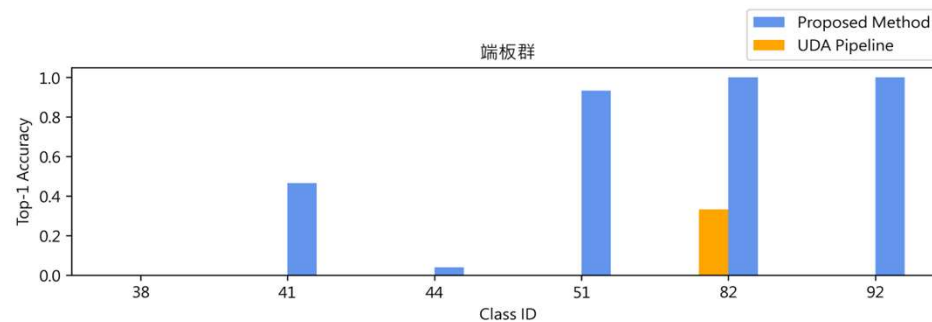
- Conclusion :
 - The result shows that our method has the advantage of recognizing the similar components, even the UDA Pipeline adding the real images in training phase.

Experiment 2 : Result

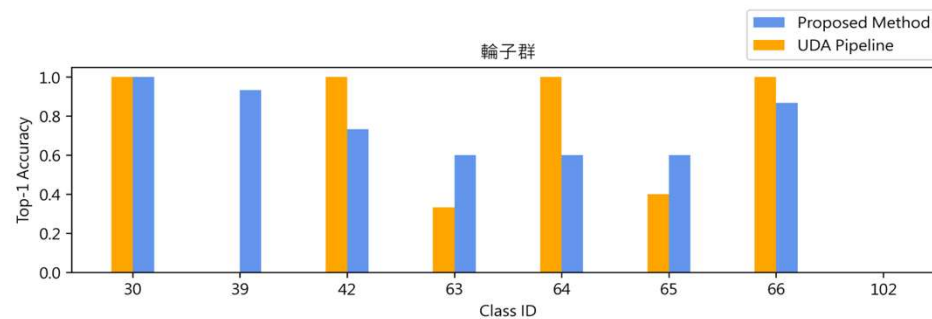
- Show the result 端板群、輪子群 individually.

Cluster	Proposed Method		UDA Pipeline	
	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc
端板群	57.33%	87.77%	5.56%	82.22%
輪子群	66.67%	86.7%	59.17%	87.50%

- Conclusion :
 - For categories with distinct features or length differences, our proposed method improves performance—even when the UDA pipeline include real images during training.



• Top1-Acc



• Top1-Acc

Classes	Real Length (m)
Class 38	0.0279
Class 41	0.0359
Class 44	0.0129
Class 51	0.0139
Class 82	0.0279
Class 92	0.0429

Classes	Real Length (m)
Class 30	0.0699
Class 39	0.0899
Class 42	0.0359
Class 63	0.1349
Class 64	0.0699
Class 65	0.0669
Class 66	0.1349
Class 102	0.0899

Experiment 2 : Result

- Similar Components Total Result (Synthetic RGB-B)

Cluster	Proposed Method		UDA Pipeline	
	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc
端板群	35.56%	74.44%	50%	72.22%
套管群	15.83%	25.83%	12.5%	37.50%
輪子群	75.33%	92.83%	58.33%	79.16%
長方兩孔群	24.44%	55.56%	55.56%	100%
Total	37.79%	62.17%	44.00%	72.22%

- Conclusion :
 - Our method did not outperform UDA Pipeline, mainly due to the following reason:
 - UDA Pipeline adopts the CDAN-MCC method
 - Our synthetic images contain strong background interference due to the lack of material textures in the 3D CAD models
 - The benefit of length information is limited

Experiment 3: topex-printer Best Accuracy

Objective	Compare our proposed method with UDA Pipeline
Description	- Evaluate accuracy results on topex-printer dataset, trained by Topex-RGB 、Topex-RGB-R 、Topex-RGB-W. - Method Comparison: - Proposed Method - Related Work (UDA Pipeline) - The main solution provided by each method : - Proposed Method : Background Removal module 、Length Estimation (Shape-based Length Estimation module + Length-based Filtering Module) - UDA Pipeline : Adding real images in training for transfer-learning (CDAN-MCC)
Metrics	Top1 accuracy 、 Top5 accuracy

	Proposed Method	UDA Pipeline
Adding real Images in training	✗	✓
Background Removal Module	✓	✗
Length Estimation	✓	✗

	Number
Synthetic Training Images (Topex-RGB)	3264
Synthetic Training Images (Topex-RGB-R)	13056
Synthetic Training Images (Topex-RGB-W)	3264
Real Images For Training.	102
Testing Images (Length + Model)	6232 (429 + 5803)

Experiment 3 : Result

- Compare proposed method and UDA Pipeline on topex-printer dataset.
- Experiment Setting :

	Proposed Method	UDA Pipeline
Adding real Images in training	✗	✓
Background Removal Module (BR)	✓	✗
Length Estimation (LE)	✓	✗

Train Dataset	Proposed Method (w/o adding real)		UDA Pipeline		Proposed Method (w/ adding real)		UDA Pipeline + BR + LE	
	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc	Top1-Acc	Top5-Acc
Topex-RGB	61.79%	84.52%	70.46%	91.65%	72.65%	91.33%	67.42%	89.00%
Topex-RGB-R	65.27%	84.86%	64.56%	89.12%	73.66%	91.15%	62.44%	86.51%
Topex-RGB-W	64.12%	85.80%	66.15%	88.62%	73.73%	91.52%	64.42%	87.32%

- Conclusion :
 - Although the accuracy may not better without adding real images, proposed method has an advantage of data availability.
 - When all settings are the same, proposed method can perform better than UDA Pipeline.

Conclusion

Conclusion

- **Integrated framework for fine-grained recognition and length estimation** to address the challenges in classifying the fine-grained industrial components and estimating the length of target components in real-world scenarios.
- Enables **recognition and length estimation without requiring background replacement**, improving real-world usability.
- Considering the geometric of 3D components, effectively **reduces errors** caused by using a single unified **length estimation approach**.
- Our experiments show that when the model is trained solely on synthetic data, it **can outperform the existing UDA Pipeline approach**, which is using real images during the training phase.

Future work

1. Length Estimation
 - RGB and Depth images captured by Microsoft HoloLens 2 have different resolutions, leading to the certain amount of errors.
 - Using more suitable device for length estimation to reduce the error value.
2. Model Recognition
 - Try to explore RNNs models that accept multiple views as sequential input.